Can Tokens Compete? Token Representations against Supervised CNN Backbones for BirdCLEF+ 2026

Working notes · DS@GT ARC at BirdCLEF+ 2026

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Audio	522.1 h
Focal recordings	344.5 h across 206 species
Soundscapes	177.6 h from 23 sites in the Pantanal
Labeled soundscapes	1.03 h, 9 of the 23 sites
Target species	234, of which 28 from soundscapes alone
Inference budget	90 min CPU on Kaggle

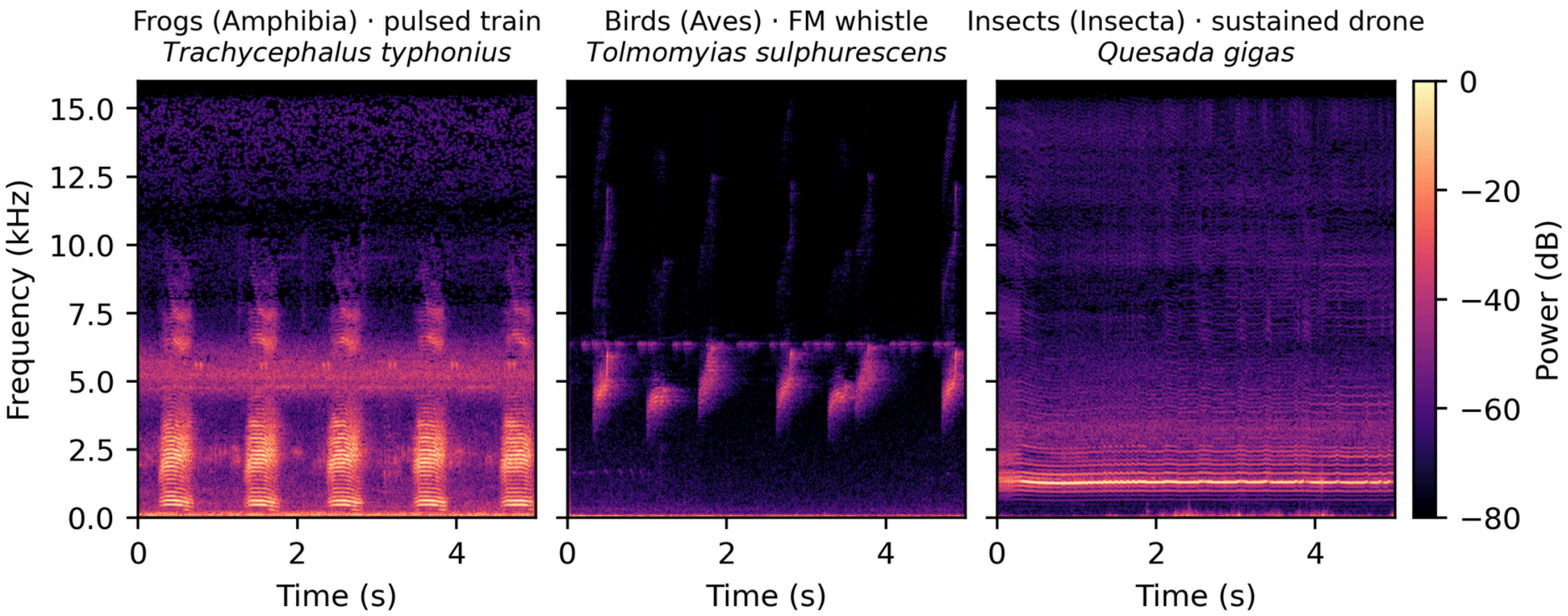
Multi-label detection on 60 s soundscapes, with about an hour of them labeled in 2026.

Class composition of focal and soundscape data

Class	Focal species	% of recordings	Soundscape species	% of windows
Aves	162	97.89	28	28.7
Amphibia	32	1.27	17	76.6
Insecta	3	0.56	25	22.7
Mammalia	8	0.28	4	5.5
Reptilia	1	0.003	1	1.8

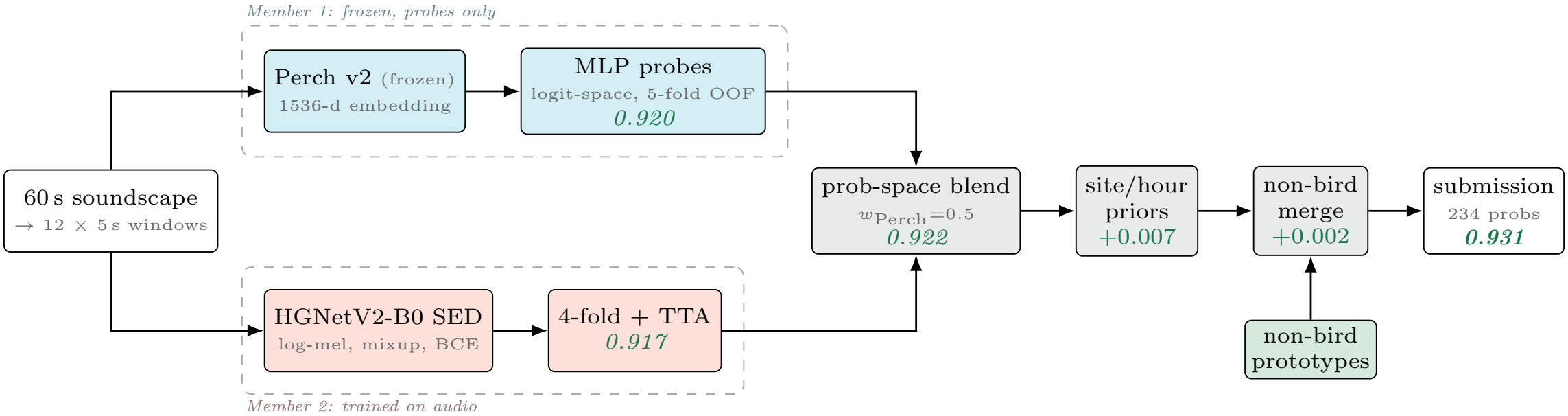
The focal training is mostly birds, while the soundscape is mostly amphibians.

Spectrograms of three vocalizing taxa



The taxa overlap in frequency and differ in spectro-temporal structure.

Supervised baseline system



Frozen Perch v2 probes and an HGNetV2-B0 SED network, blended equally. Public LB in green.

Non-bird prototypical head

$$\hat{y}_{t,s}^{\text{proto}} = \text{clip}(\cos(\boldsymbol{e}_t, \boldsymbol{\mu}_s^+) - \cos(\boldsymbol{e}_t, \boldsymbol{\mu}_s^-), 0, 1)$$

$$\hat{y}_{t,s}^{\text{final}} = \max\left(\hat{y}_{t,s}^{\text{sed}}, \hat{y}_{t,s}^{\text{proto}}\right)$$

Closeness to windows with the species minus closeness to windows without it, in Perch space. Macro AP on non-bird taxa 0.895 against 0.074 for the SED alone.

Baseline ablations on the Kaggle leaderboard

Configuration	public	private	Δ
Members, standalone			
Perch v2 + ridge probe, frozen	0.866	0.862	
Perch v2 + MLP probes, frozen	0.920	0.918	+0.054
HGNetV2-B0 SED, 4-fold	0.917	0.924	+0.051
Ensemble, $w_{\text{Perch}} = 0.5$			
with ProtoSSM head	0.927	0.923	+0.061
MLP-only, drop ProtoSSM	0.929	0.925	+0.063
no post-processing	0.922	0.916	+0.056
MLP-only + non-bird prototypical head	**0.931**	**0.936**	+0.065

Δ is the public gain over the frozen ridge probe. The final ensemble is 0.936 private at rank 1894.

Token representations for bioacoustic soundscapes

	Codec representations	Semantic representations
Source	neural audio codecs	foundational audio embeddings
Trained to	reconstruct the waveform	preserve similarity and separability
Pretraining	human speech	AudioSet
Encoders	WavTokenizer	AST, BEATs, EAT, SSAMBA

We rule out the reconstruction route, then turn to the semantic embeddings.

WavTokenizer codec retrieval against its success gates

Phase	Metric	Target	Result	Met?
Round-trip	Perch embedding cosine	> 0.90	0.313	×
Round-trip	Precision@1	> 0.80	0.09	×
Focal retrieval	Precision@10	> 0.80	0.195	×

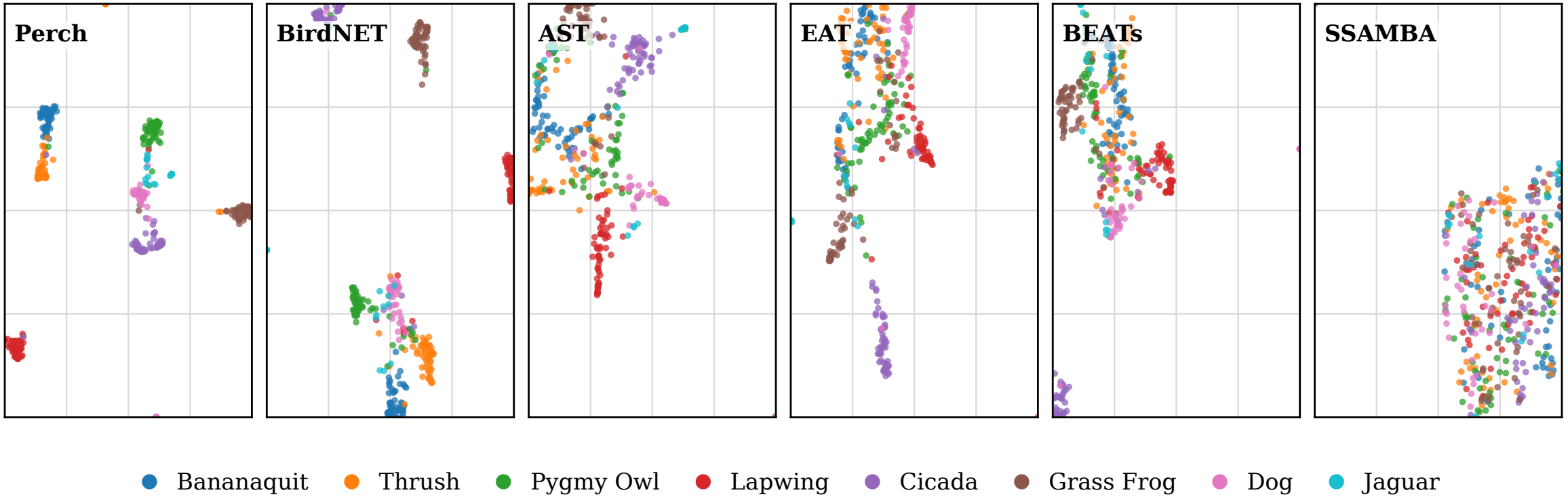
Perch v2 is the round-trip evaluator. Every gate is missed by a wide margin.

Frozen LOSO ranking versus the Kaggle leaderboard

Encoder	dim	LOSO AUROC	public	private	CPU min/1k
Perch	1536	**0.620**	**0.866**	**0.862**	**11.5**
BEATs	768	0.612	0.777	0.760	29.6
BirdNET	1024	0.588	0.814	0.798	34.5
AST	768	0.576	0.778	0.768	38.2
EAT	768	0.569	0.780	0.766	25.5
SSAMBA	384	0.563	0.731	0.689	72.5

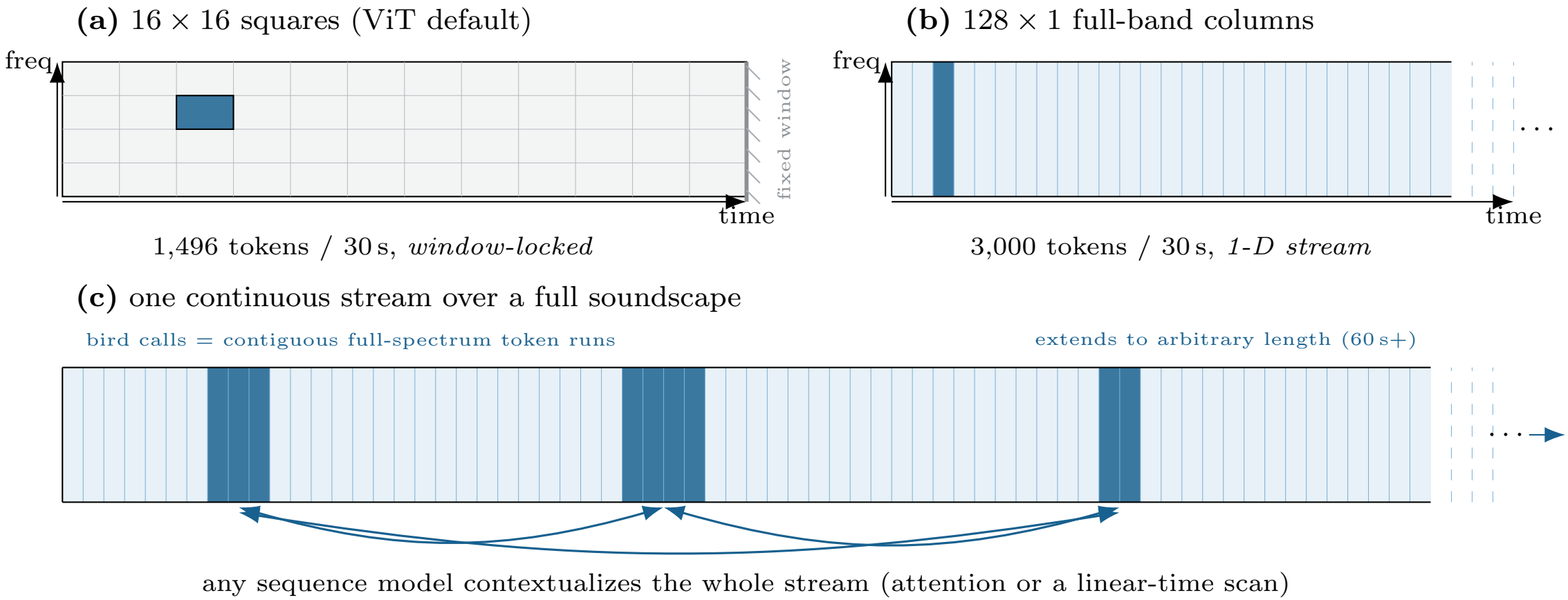
Same ridge head offline and on Kaggle. The offline ranking does not predict the leaderboard.

Embedding-space species separability



PaCMAP of frozen embeddings for eight species. The specialists are nearly linearly separable.

Toward sequence modeling of arbitrary-length audio



Full-band 128 × 1 columns give a one-dimensional token stream over an entire soundscape.

Summary

Finding	Evidence
The supervised baseline carried the result	Private LB 0.936, rank 1894
Codec representations fail bioacoustic retrieval	Round-trip cosine 0.313, gate 0.90
Semantic embeddings transfer modestly	Public LB 0.777 for BEATs, 0.866 for Perch

The linear-time state-space alternative remained inconclusive at the scale we could train.

Conclusion

We look forward to foundational bioacoustic systems that exploit unlabeled data and make use of low-resource classes, in a domain where expertise is limited and conservation needs are urgent.

Code: dsgt-arc/birdclef-2026